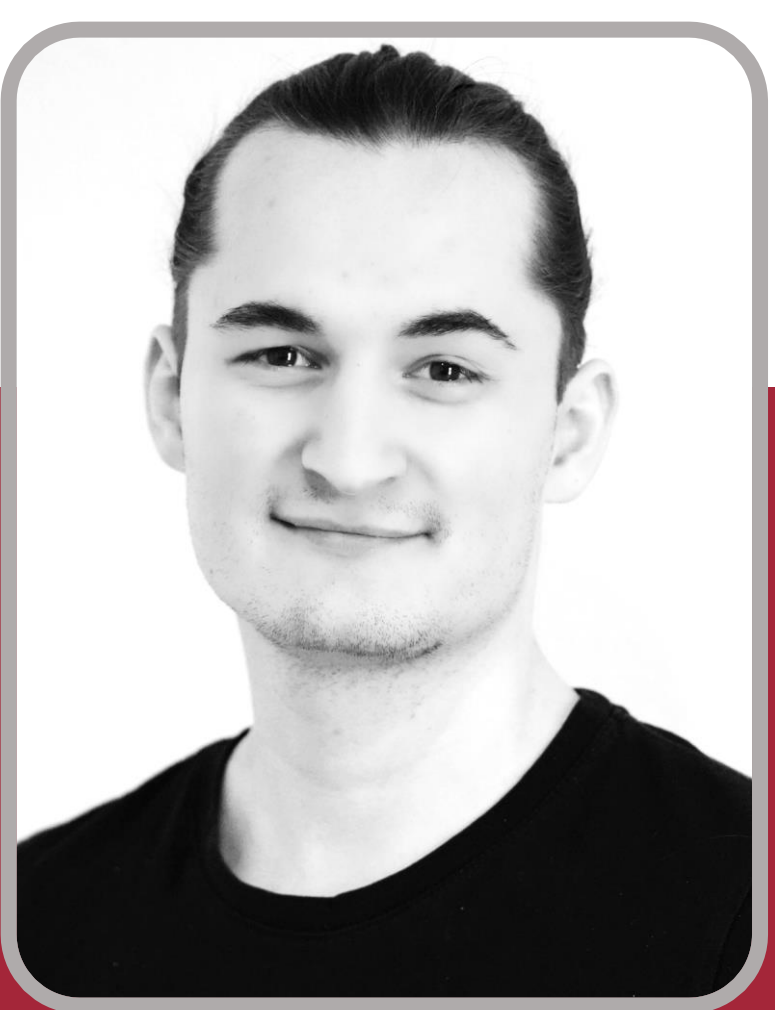




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Toward Multi-Modal In-Situ Design for Personal Fabrication through Generative AI and Shared Representations

Enhancing Quality Assurance in Additive Manufacturing through Augmented Reality

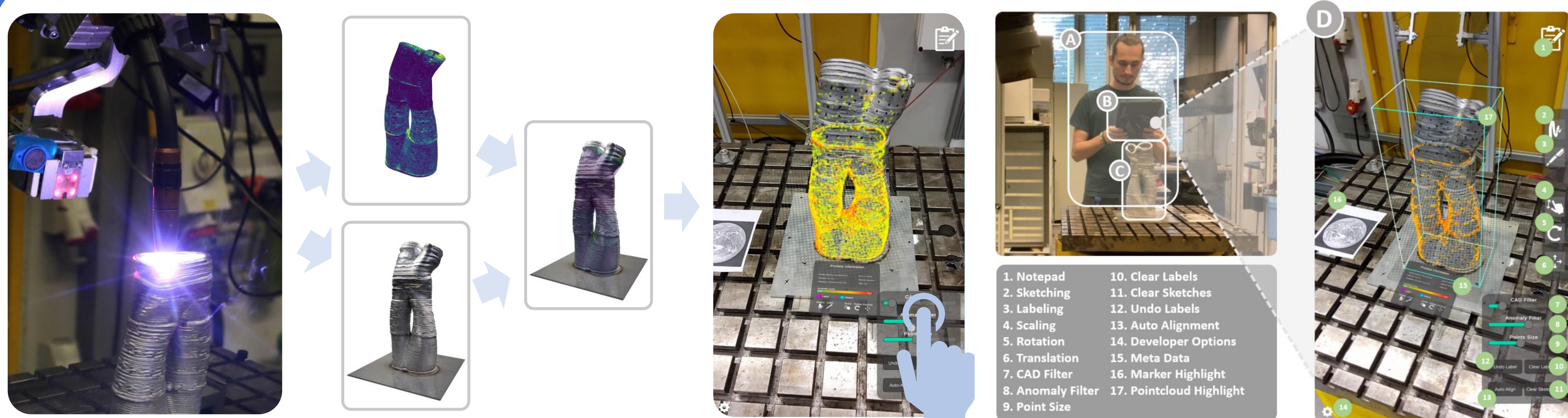


Figure: AR prototype for in-situ visualization of data from manufacturing processes. From left to right: Manufacturing process, data overlay, system setup, user interface.

Additive Manufacturing (AM) is a pivotal technology for future fabrication. However, Quality Assurance (QA) remains a challenge due to the stochastic nature of physical processes. To address this, we apply **Augmented Reality (AR)** by visualizing spatio-temporal **Digital Twin** data and providing **intuitive interactions for effective data labelling**.

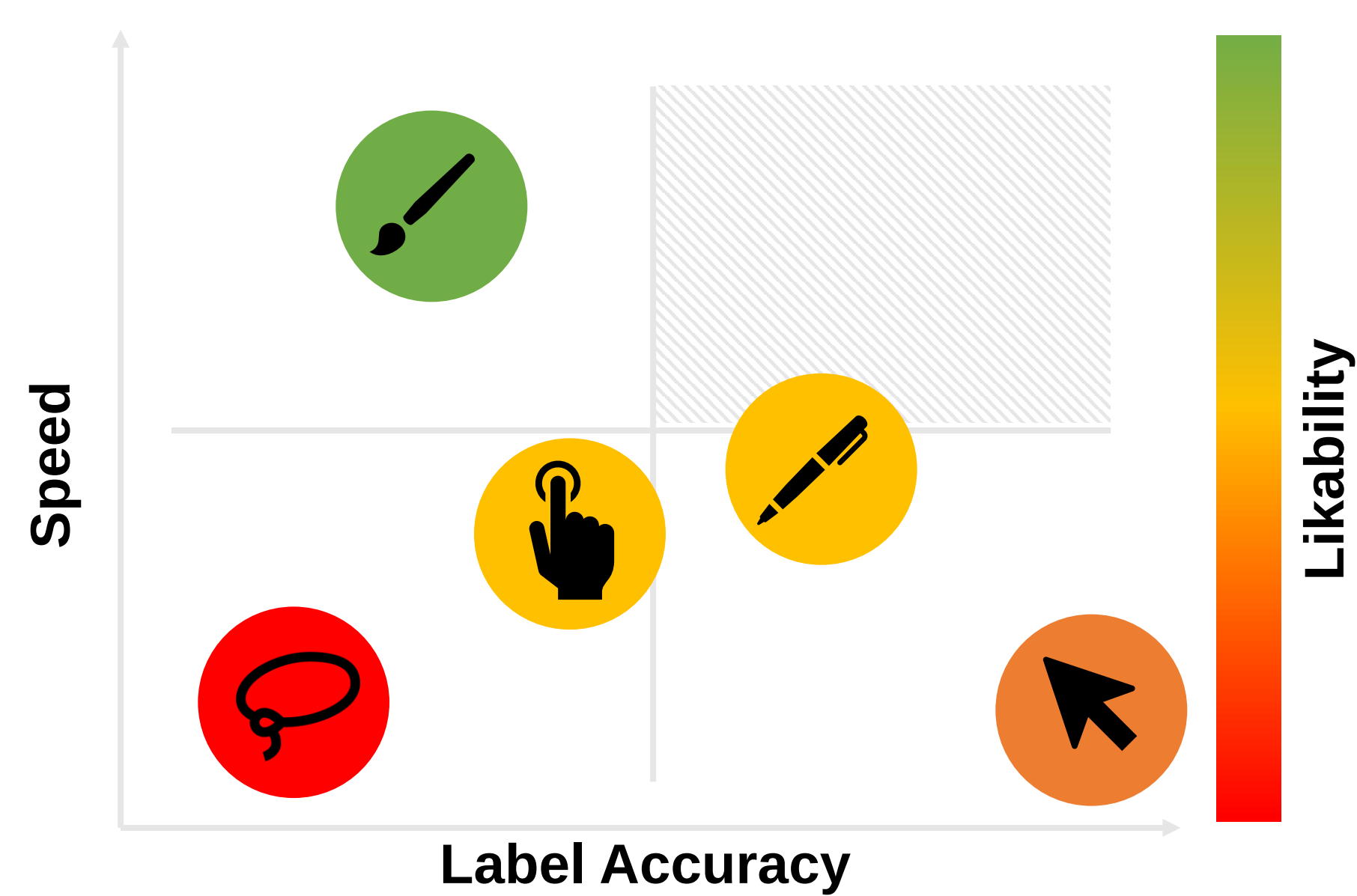


Figure: trade-offs for developed labelling methods.

VRSketchIn: Exploring the Design Space of Pen and Tablet Interaction in VR

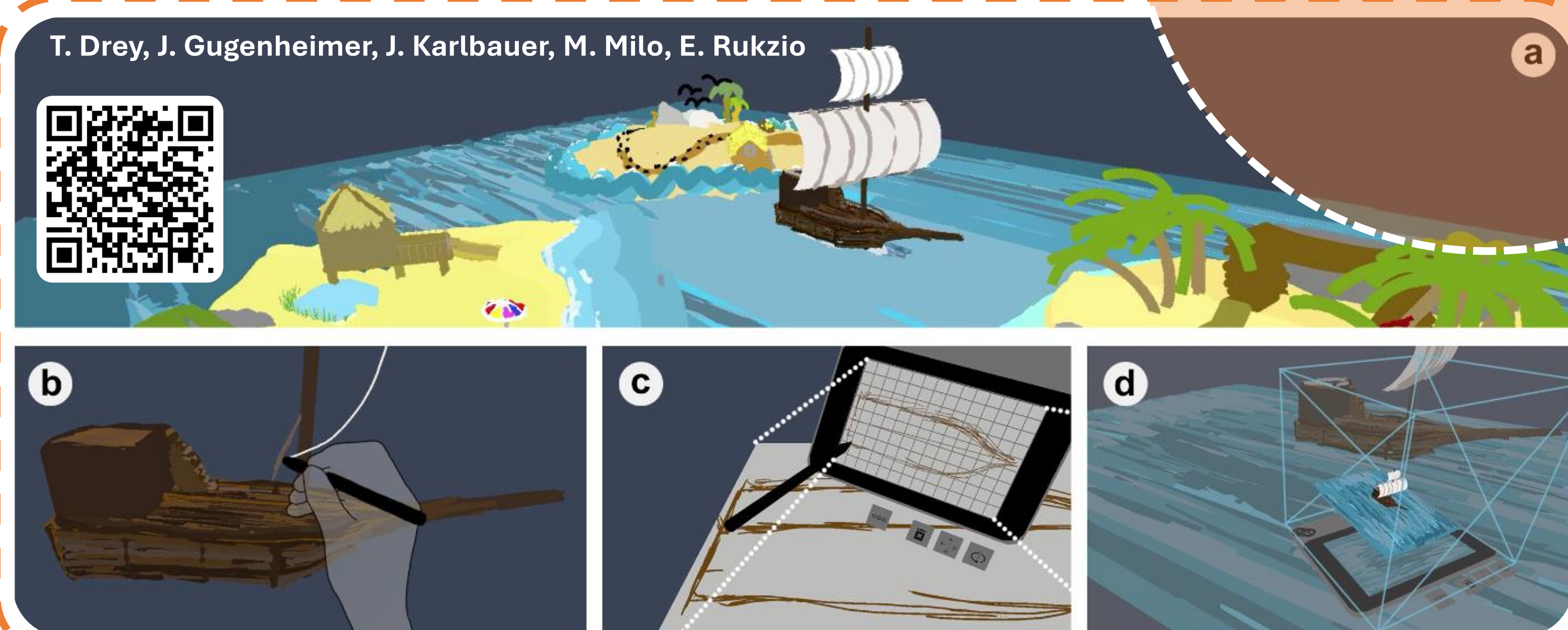
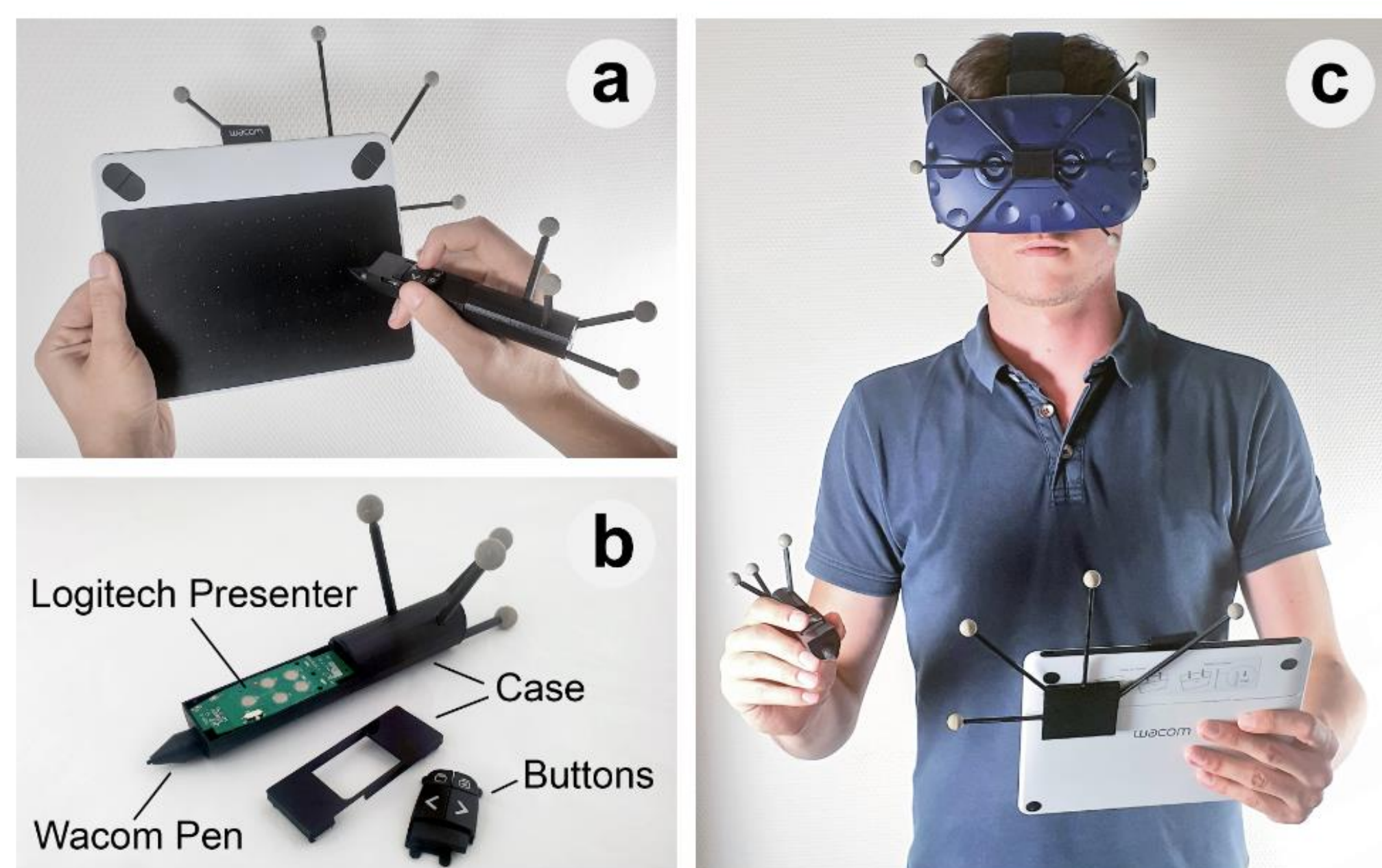


Figure: VR sketching prototype. **a)** an overview of user generated content. **b)** sketching in mid-air. **c)** sketching on tablet. **d)** a demonstration of the world-in-miniature tool.

VRSketchIn is a **VR sketching application combining 3D with 2D sketching**. We explore the possibilities arising from combining 2D and 3D input and define the design space for sketching metaphors. To understand sketching operations with 2D and 3D metaphors, we present findings of (N = 6) usability walkthroughs.



Raw Audio Music Emotion Composition Using Mood Annotated Spotify Playlists

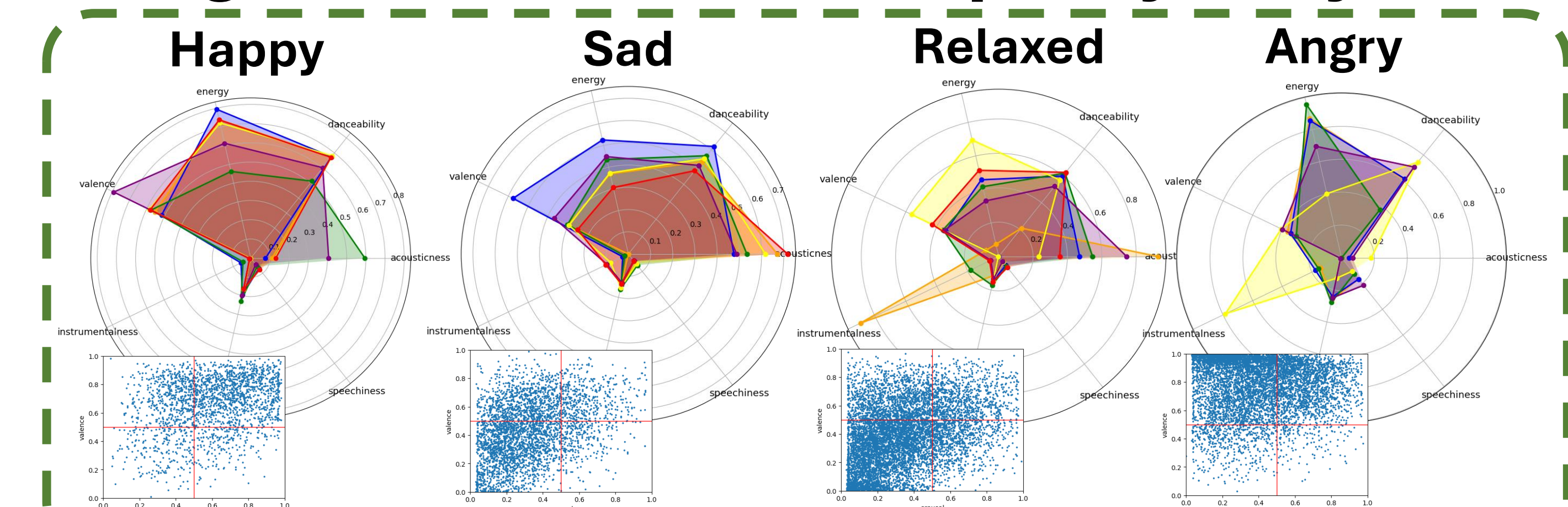


Figure: Radar plots show feature distributions of annotated playlists. Scatter plots show the distribution of music samples on a valence-arousal plane.

The lack of annotated data poses a challenge in emotion expression for supervised learning methods. Using **crowd-annotated playlists** from Spotify, we trained a **transformer model for generating music** for quadrants in a valence-arousal plane. Experiments demonstrate the model's ability to **effectively mimic mood present in given datasets**.

Digital
Fabrication

Personal
Fabrication

GenAI

Mixed
Reality

Future Work

Toward Multi-Modal In-Situ Design for Personal Fabrication through Generative AI and Shared Representations

Generative AI for 3D modeling addresses two prominent challenges in **Personal Fabrication**. First, it yields the potential to reduce the required time to create CAD models by orders of magnitude, and second, it opens the space of CAD to a broad mass of novice users with little to no knowledge of 3D modeling software. However, AI prompting is ambiguous and enforces specific prompting modalities through underlying AI architectures. In my work I want to **enable multi-modal Generative Design in Augmented Reality to enhance Personal Fabrication**.

Augmented Reality and Digital Fabrication share the motivation to bring digital and physical worlds closer together through **Shared Representation Spaces**. A similar development can be observed in **Generative AI** through emerging **Shared Feature Spaces**. In my work I want to **combine Shared Representations in AR and Gen AI for Personal Fabrication**.